# Yuqing Wang

ywan1050@jh.edu
https://yzwangyuqing.github.io

## RESEARCH INTERESTS

machine learning theory, optimization, (stochastic) dynamical systems, sampling, computational mathematics

## PROFESSIONAL EXPERIENCE

2025 - Postdoctoral Fellow
Department of Applied Mathematics and Statistics
Johns Hopkins University

Fall 2024 Research Fellow
Program: *Modern Paradigms in Generalization*, joint with *Special Year on Large Language Models and Transformers Part 1*
Simons Institute for the Theory of Computing, University of California, Berkeley

## EDUCATION

Aug 2024 Ph.D. in Mathematics
Georgia Institute of Technology
Advisor: Prof. Molei Tao

June 2018 B.S. in Mathematics
Nankai University

## PREPRINTS AND PUBLICATIONS

**Yuqing Wang**, Shangding Gu. "Data Uniformity Improves Training Efficiency and More, with a Convergence Framework Beyond the NTK Regime". preprint.

**Yuqing Wang**, Ye He, Molei Tao. "Evaluating the design space of diffusion-based generative models". Advances in Neural Information Processing Systems 37 (NeurIPS 2024).

Zhenghao Xu, **Yuqing Wang**, Tuo Zhao, Rachel Ward, Molei Tao. "Provable Acceleration of Nesterov's Accelerated Gradient for Rectangular Matrix Factorization and Linear Neural Networks". Advances in Neural Information Processing Systems 37 (NeurIPS 2024).

**Yuqing Wang**, Zhenghao Xu, Tuo Zhao, Molei Tao. "Good regularity creates large learning rate implicit biases: edge of stability, balancing, and catapult". Accepted (JMLR).

Bo Yuan, Jiaojiao Fan, **Yuqing Wang**, Molei Tao, and Yongxin Chen. "Markov Chain Monte Carlo for Gaussian: A Linear Control Perspective", in IEEE Control Systems Letters, vol. 7, pp. 2173-2178, 2023, doi: 10.1109/LCSYS.2023.3285140.

Lingkai Kong, **Yuqing Wang**, and Molei Tao. "Momentum Stiefel Optimizer, with Applications to Suitably-Orthogonal Attention, and Optimal Transport". International Conference on Learning Representations (ICLR 2023).

**Yuqing Wang**, Minshuo Chen, Tuo Zhao and Molei Tao. "Large Learning Rate Tames Homogeneity: Convergence and Balancing Effect". International Conference on Learning Representations (ICLR 2022).

Kaixuan Huang*, **Yuqing Wang***, Molei Tao, and Tuo Zhao."Why Do Deep Residual Networks Generalize Better than Deep Feedforward Networks?—A Neural Tangent Kernel Perspective." Advances in neural information processing systems 33 (NeurIPS 2020): 2698-2709. (*: equal contribution)

## TALKS

| | |
|---|---|
| Jan 2026 | *Data uniformity improves training efficiency and more, with a convergence framework beyond the NTK regime*, AMS Special Session on Algebraic Methods in Machine Learning and Optimization, Joint Mathematics Meetings (JMM 2026), Washington, D.C. |
| Nov 2025 | *Data uniformity improves training efficiency and more, with a convergence framework beyond the NTK regime*, Scientific Machine Learning with Robust Computation Minisymposium, SIAM New York-New Jersey-Pennsylvania Section Conference, Pennsylvania State University |
| July 2025 | *Theoretical implications of training and sampling diffusion models*, "Optimization Meets GenAI", International Conference on Continuous Optimization (ICCOPT), University of Southern California |
| May 2025 | *The mechanism behind the implicit biases of large learning rates: edge of stability, balancing, and catapult*, Math Machine Learning seminar, Max Planck Institute for Mathematics in the Sciences + University of California, Los Angeles |
| May 2025 | *Theoretical implications of training and sampling diffusion models*, Dynamical Systems for Machine Learning (MS141), SIAM Conference on Applications of Dynamical Systems (DS25), Denver, Colorado |
| Apr 2025 | *The mechanism behind the implicit biases of large learning rates: edge of stability, balancing, and catapult*, Computational Mathematics + X (CMX) seminar, California Institute of Technology |
| Apr 2025 | *The mechanism behind the implicit biases of large learning rates: edge of stability, balancing, and catapult*, Applied Mathematics and Statistics (AMS) postdoc seminar, Johns Hopkins University |

| | | |
|---|---|---|
| Oct | 2024 | *The mechanism behind the implicit biases of large learning rates: edge of stability, balancing, and catapult*, SIAM Conference on Mathematics of Data Science (MDS24), October 21–25, 2024, Atlanta, Georgia |
| Sep | 2024 | *How well does diffusion model generate?—A training and sampling combined quantification*, Emerging Generalization Settings Workshop, Simons Institute, University of California, Berkeley (joint with Molei Tao) |
| Apr | 2024 | *What creates edge of stability, balancing, and catapult*, the Second Southeast Applied and Computational Math Student Workshop, Georgia Institute of Technology |
| Dec | 2023 | *Quantitative acceleration of convergence to invariant distribution by irreversibility in diffusion processes*, PDE seminar, Georgia Institute of Technology |
| Oct | 2023 | *What creates edge of stability, balancing, and catapult*, ACO student seminar, Georgia Institute of Technology |
| May | 2023 | *Implicit bias of large learning rate*, SIAM Conference on Applications of Dynamical Systems (DS23), Machine Learning for Dynamical Systems & Dynamical Systems for Machine Learning - Part I of III (MS143), Portland, Oregon |

## HONORS

| | |
|---|---|
| 2025 | *Sigma Xi Best Ph.D. Thesis Award*, Georgia Institute of Technology |
| 2024 | *Outstanding Teaching Assistant*, Georgia Institute of Technology |
| 2023 | *Rising Star in Data Science*, University of Chicago & University of California San Diego |
| 2023 | *Rising Star in Electrical Engineering and Computer Sciences (EECS)*, Georgia Institute of Technology |
| 2023 | *Top Graduate Student*, Georgia Institute of Technology |
| 2022 | *"Thank a teacher" Award*, Georgia Institute of Technology |

## SERVICE

*Organizer:*

| | | |
|---|---|---|
| May | 2025 | Minisymposium on "Dynamical Systems for Machine Learning", SIAM Conference on Applications of Dynamical Systems (DS25), Denver, Colorado |
| 2025 - | | Applied Mathematics and Statistics postdoc seminar, Johns Hopkins University |
| Fall | 2024 | Modern Paradigms in Generalization/Large Langnguage Models program seminar, Simons Institute, University of California, Berkeley |
| Fall | 2024 | Women+ mentoring sessions, Simons Institute, University of California, Berkeley |

Oct 2024 Minisymposium on “The dynamical view of machine learning”, SIAM Conference on Mathematics of Data Science (MDS24), Atlanta, Georgia

Fall 2021 SIAM student seminar, Georgia Institute of Technology

*Reviewer:*

NeurIPS, ICLR, Nature Communications, Journal of Machine Learning Research (JMLR)

*Others:*

Fall 2021 Vice President for SIAM Student Chapter, Georgia Institute of Technology

## TEACHING

*Instructor* in the Department of Applied Mathematics and Statistics, Johns Hopkins University:

Fall 2025
EN 553.432 Bayesian Statistics

Spring 2025
EN 553.361 Introduction to Optimization I

*Teaching Assistant* in School of Mathematics, Georgia Institute of Technology:
*teach 4 hours of studios per week*

Fall 2018, Spring 2019, Fall 2019, Fall 2020, Spring 2021, Fall 2023
MATH 2552 Differential Equation

Spring 2020
MATH 2551 Multivariable Calculus

Summer 2019, Summer 2024
MATH 2550 Introduction to Multivariable Calculus

*Head Teaching Assistant* in the School of Mathematics, Georgia Institute of Technology:
*design worksheets; teach 2 hours of studios per week*

Spring 2022, Spring 2023
MATH 2552 Differential Equation